FIITJEE **JEE (Advanced)-2018**

ALL INDIA TEST SERIES

FULL TEST – IV

Paper 2

Time Allotted: 3 Hours **Maximum Marks: 240**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C.**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

1. **Section–A (01 – 08, 21 – 28, 41 – 48)** contains 24 multiple choice questions which have one or more correct answer. Each question carries **+4 marks** for correct answer and **–2 marks** for wrong answer.

   Section–A (09 – 12, 29 – 32, 49 – 52) contains 12 paragraphs with each having 2 questions with one correct answer. Each question carries **+4 marks** for correct answer and **–2 marks** for wrong answer.

2. **Section–C (13 – 20, 33 – 40, 53 – 60)** contains 24 Numerical based questions with answers as numerical value from 0 to 9 and each question carries +4 marks for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

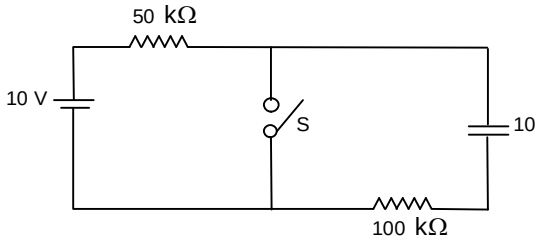
PART – I

SECTION – A
One OR More Than One Choice Type

This section contains **8 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct.

1. The equation of a transverse wave travelling along a string is given by $y = 0.4\sin\left[\pi(0.5x - 200t)\right]$ where y and x are measured in cm and t in seconds. Suppose that you clamp the string at two points L cm apart and you observe a standing wave of the same wavelength as above transverse wave. For what values of L less than 10 cm is this possible?
(A) 2 cm (B) 4 cm
(C) 9 cm (D) 8 cm

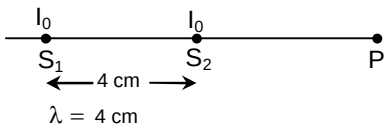
2. In the circuit shown, the switch S has been closed for a long time. At time t = 0, the switch is opened. It remains open for a long time T, after which it is closed again.



(A) voltage drop across 100 k Ω resister is $10e^{-t/1.5}$ V for t < T
(B) voltage drop across 100 k Ω resister is $\frac{20\,e^{-t/1.5}}{3}$ V for t < T
(C) voltage drop across 100 k Ω resister is $100\left(e^{T-t} - e^{(T-3t)/3}\right)$ for t > T
(D) The time constant of the circuit is 1.5 sec for t > T

Space for Rough work

3. Two coherent sources S_1 and S_2 are situated as shown in figure. Here S_1 is fixed but S_2 is performing SHM with amplitude 2 cm. Initially S_2 is at right extreme. At time $t = \frac{\pi}{2\omega}$ (where ω = angular frequency of SHM):



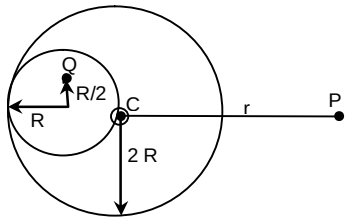
(A) Phase difference between waves produce by S_1 and S_2 at point P is $\frac{\pi}{2}$.

(B) Intensity at point P is $2I_0$.

(C) Intensity at point P is $4I_0$

(D) Intensity at point P is zero

4. There is a conductor with cavity of radius R as shown in figure. A point charge Q is placed at R/2 distance from centre of cavity. A point P is at r distance from centre of conductor. Then,



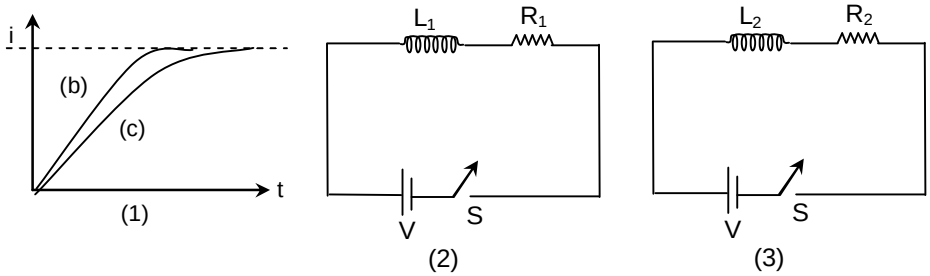
(A) charge density at inner surface of cavity will be non uniform

(B) charge density at surface of conductor will be uniform

(C) electric field at point P is $\frac{Q}{4\pi\varepsilon_0 r^2}$

(D) force on charge Q is non zero

5. Current growth in two L – R circuits (2) and (3) is shown in figure (1) after switch S closed. Then



(A) $R_1 > R_2$

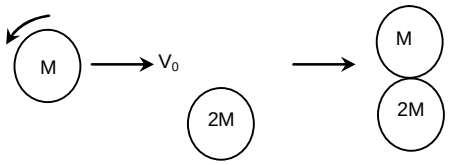
(B) $R_1 = R_2$

(C) $L_1 > L_2$

(D) $L_1 < L_2$

Space for Rough work

6. A uniform disc of mass m and diameter 2 R moves forward towards another uniform disc of mass 2 m & diameter 2 R on a frictionless surface as shown in figure



When the first disc contacts the second, they stick to each other and move as a single object.

(A) the velocity of combined disc after the collision is $\frac{V_0}{3}$.

(B) the angular velocity of combined disc after the collision is $\frac{\omega_0}{3}$.

(C) If $\omega_0 = \frac{8}{3}\frac{V_0}{R}$, the combined disc will not rotate

(D) If combined disc does not rotate, the energy loss is $\frac{19}{9}MV_0^2$.

7. Two strings with linear mass densities $\mu_1 = 0.1$kg/m and $\mu_2 = 0.3$ kg/m are joined seamlessly. They are under tension of 20 N. A travelling wave of triangular shape is moving from lighter to heavier string.
(A) The reflection coefficient at interface is zero
(B) The reflection coefficient at interface is $2-\sqrt{3}$
(C) The transmission coefficient at interface is 1
(D) The transmission coefficient at interface is $\sqrt{3}-1$

8. Consider a piston – cylinder arrangement with spring in its natural length. Due to heat transfer, the gas expands until the piston hits the stop. The mass of piston is 10 kg and area is 78 cm^2. Initial and final specific internal energies are 214 and 337 k J / kg.
(A) Initial pressure of gas is 112 k Pa
(B) Work done by gas on the piston is 56.5 J
(C) Work done by gas on the piston is 100 J
(D) Heat transfer for 0.5 gas is 118 J

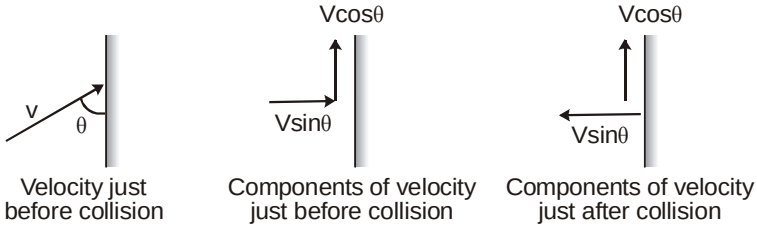
Space for Rough work

Comprehension Type

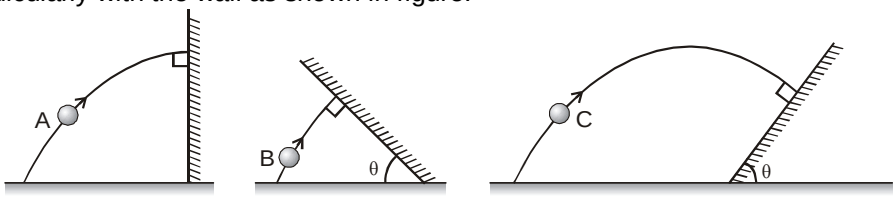
This section contains **2 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct.

Paragraph for Question Nos. 09 and 10

We know how by neglecting the air resistance, the problems of projectile motion can be easily solved and analysed. Now we consider the case of the collision of a ball with a wall. In this case the problem of collision can be simplified by considering the case of elastic collision only. When a ball collides with a wall we can divide its velocity into two components, one perpendicular to the wall and other parallel to the wall. If the collision is elastic then the perpendicular component of velocity of the ball gets reversed with the same magnitude.



The other parallel component of velocity will remain constant if wall is given smooth.
Now, let us take a problem. Three balls A, B and C are projected from ground with same speed at same angle with the horizontal. The ball A, B and C collide with the wall during their flight in air and all three collide perpendicularly with the wall as shown in figure.



9. If the total time taken by the ball A to fall back on ground is 4 seconds and that by ball B is 2 seconds. Then, the maximum height attained by ball B from ground will be
(A) 15 m (B) 5 m
(C) 3 m (D) 25 m

Space for Rough work

10. The vertical component of velocity of balls with which they are projected is
(A) 20 m/s (B) 10 m/s
(C) $10\sqrt{3}$ m/s (D) Undeterminable

Paragraph for Question Nos. 11 and 12

We have a disc of negligible thickness and whose surface mass density varies as radial distance from centre as $\sigma = \sigma_0\left(1+\frac{r}{R}\right)$, where R is the radius of the disc. Specific heat of the material of the disc is C. Disc is given an angular velocity ω_0 and placed on a horizontal rough surface such that the plane of the disc is parallel to the surface. Coefficient of friction between disc and surface is μ. The temperature of the disc is T_0.
Answer the following 3 questions on the base of information provided in the above paragraph.

11. Magnitude of angular acceleration of disc is
(A) $\alpha = \frac{7\mu g}{27R}$ (B) $\alpha = \frac{21\mu g}{27R}$
(C) $\alpha = \frac{35\mu g}{27R}$ (D) $\alpha = \frac{42\mu g}{27R}$

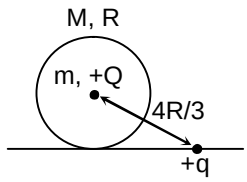
12. If the kinetic energy lost due to the friction between disc and surface is absorbed by the disc the moment heat is generated, by the mass lying at the point where the energy is lost. Assume that there is no radial heat flow in between disc particles. Rate of change in temperature at a point lying at r distance away from centre with time, before disc stops rotating, will be (α - magnitude of angular acceleration of rod)
(A) $\frac{\mu gr}{C}(\omega_0 - \alpha t)$ (B) $\frac{\mu gr}{2C}(\omega_0 - \alpha t)$
(C) $\frac{\mu gr}{3C}(\omega_0 - \alpha t)$ (D) $\frac{\mu gr}{4C}(\omega_0 - \alpha t)$

Space for Rough work

SECTION – C
(One Integer Value Correct Type)

This section contains **8 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

13. There is a disc of mass M and radius R. A point charge +Q and of mass m is fixed at the centre of the disc. The disc is held on a fixed horizontal rough surface. Another point charged +q is fixed on the surface such that distance between +q and +Q is $\frac{4R}{3}$. Now the disc is set free. If the disc does not loose contact with the ground at the same intent the relation must hold is $M > \eta m$. Find η. (Use $\frac{27Qq}{256\pi\varepsilon_0 R^2} = 3mg$)



14. Two identical containers contain same amount of ideal gas at identical condition. Gas in one container is expanded to 4 times of its volume at constant pressure and gas in another container is expanded to four times of its volume by isothermal process. If W_1 and W_2 be the respective work done, then $\frac{W_1}{W_2} = \frac{\alpha}{\ln 2}$. Find $\frac{2}{3}\alpha$.

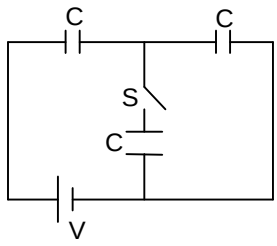
15. A hydrogen atom moving at a speed v_0 absorbs a photon of wavelength 61 nm and stops. Mass of the hydrogen atom is 1.67 x 10^{-27} kg. When calculated v_0 comes out to be $\frac{13}{n}$ m/s. Find n nearly.

16. An inductor coil stores 16 joules of energy and dissipates energy as heat at the rate of 320 W when a current of 2 Amp. is passed through it. When the coil is joined across a battery of emf 20 V and internal resistance 20Ω, the time constant for the circuit is τ. Find 100τ (in seconds).

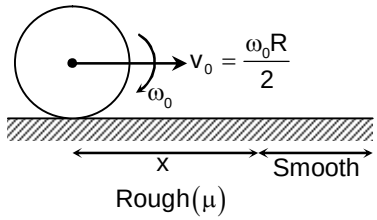
17. ABCDEF is a regular hexagon of side a. At each vertex A, B and C a point charge +q is fixed and D, E, F a point charge –q is fixed. Intensity of electric field at a point lying on the line joining FC at a distance x away from the centre of hexagon (x >>>> a) is found to be $\frac{\sqrt{\alpha}aq}{2\pi\varepsilon_0 x^3}$, where α is an integer. Find α.

Space for Rough work

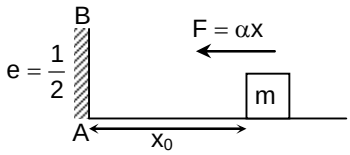
18. Initially switch S is opened and energy stored between the plates of capacitor is E. Now the switch S is closed. Work done by battery after the switch S is closed is W. Find $\frac{2E}{W}$.



19. A uniform disc of mass m and radius R is given a velocity v_0 and angular velocity ω_0 as shown and released of a fixed horizontal surface. Only the x length of the surface is rough. Find minimum value of x (in metres) so that disc starts pure rolling. Use $\frac{\omega_0^2 R^2}{72\mu g} = 1\,\text{m}$.



20. Consider a hypothetical field in which force acting on a particle varies with its horizontal distance from wall AB is $F = \alpha x$ and always towards wall. Coefficient of restitution for collision between particle and wall is $e = \frac{1}{2}$. Find the maximum force (in N) acting on the particle after 3rd collision. (use αx_0 = 16 N)



Space for Rough work

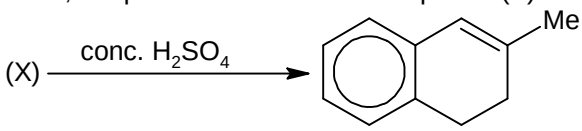
Chemistry

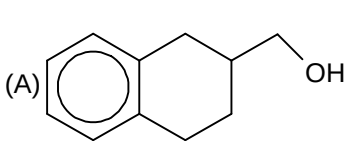
PART – II

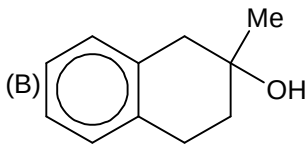
SECTION – A
One OR More Than One Choice Type

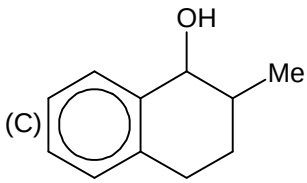
This section contains **8 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct.

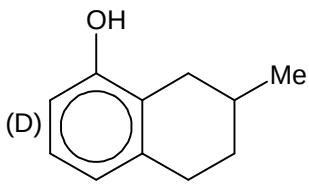
21. In the given reaction, the possible structure of compound (X) is











22. Which of the following reaction will give an alkyne?

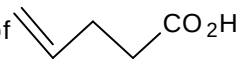
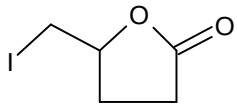
(A) Potassium fumarate $\xrightarrow{\text{Electrolysis}}$

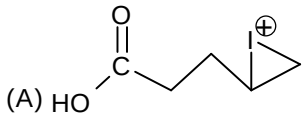
(B) $CH_3CBr_2CHBr_2 \xrightarrow{\text{Zn dust}}$

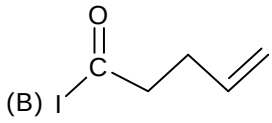
(C) $CH_3CH_2CHBr_2 \xrightarrow{\text{alc.KOH/NaNH}_2/\Delta}$

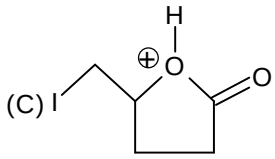
(D) $CH_3CHBrCH_2Br \xrightarrow{\text{NaNH}_2/\Delta}$

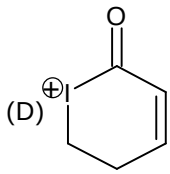
Space for Rough work

23. The iodo lactisation of  with I_2 gives the compound  + HI. The different intermeidates are

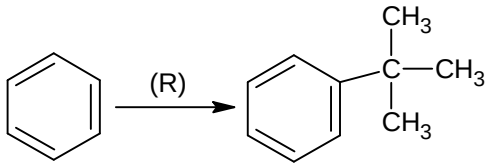
(A) 

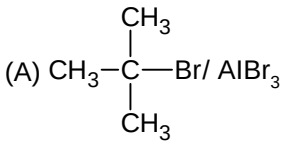
(B) 

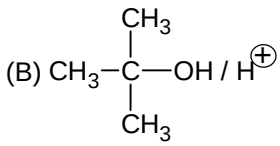
(C) 

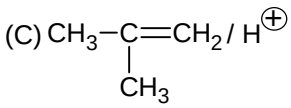
(D) 

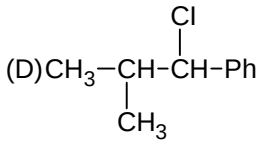
24. For the given reaction

 (R) will be

(A) 

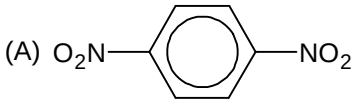
(B) 

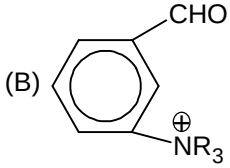
(C) 

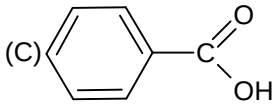
(D) 

Space for Rough work

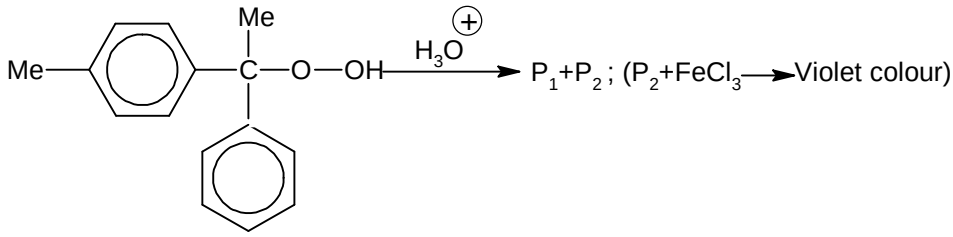
25. Which of the following compound will not give Friedel-Crafts reaction?

(A) 

(B) 

(C) 

(D) C_5H_5N

26. 

$P_1 + NaOI \rightarrow P_3 \downarrow + P_4$

Yellow

Correct statement for the above sequence is

(A) P_3 on reaction with Ag gives acetylene

(B) P_4 on reaction with sodalime gives toluene

(C) P_4 on reaction with sodalime gives benzene

(D) P_1 on reaction with 2,4-DNP gives yellow compound

27. The state of hybridization in the anionic Part of solid Cl_2O_6 is

(A) sp^3d^2 (B) sp^3d

(C) sp^3 (D) sp^2

28. In the reaction: $3Br_2 + 6CO_3^{2-} + 3H_2O \rightarrow 5Br^- + BrO_3^- + 6HCO_3^-$

(A) Bromine is oxidised and carbonate is reduced

(B) Bromine is oxidised

(C) Bromine is reduced

(D) It is disproportionation reaction or autoredox change

Space for Rough work

Comprehension Type

This section contains **2 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct.

Paragraph for Question Nos. 29 and 30

In iodometric titrations, an oxidizing agent such as $KMnO_4$, $K_2Cr_2O_7$, $CuSO_4$, H_2O_2 is allowed to react in neutral medium or in acidic medium with excess of potassium iodide to liberate free iodine KI+ oxidizing agent $\rightarrow I_2$

Free iodine is titrated against standard reducing agent usually with sodium thiosulphate i.e.,

$$K_2Cr_2O_7 + 6KI + 7H_2SO_4 \longrightarrow Cr_2(SO_4)_3 + 4K_2SO_4 + 7H_2O + I_2$$

$$2CuSO_4 + 4KI \longrightarrow Cu_2I_2 + 2K_2SO_4 + I_2$$

$$I_2 + Na_2S_2O_3 \longrightarrow 2NaI + Na_2S_4O_6$$

In iodometric titrations, starch solution is used as an indicator. Starch solution gives blue or violet colour with free iodine. At the end point, blue or violet colour disappears when iodine is completely changed to iodide.

29. What volume of 0.40 M $Na_2S_2O_3$ would be required to reach with I_2 liberated by adding 0.04 mole of KI to 50 mL of 0.20 M $CuSO_4$ solution?
(A) 12.5 mL (B) 25 mL
(C) 50 mL (D) 2.5 mL

30. A 1.1 g sample of copper ore is dissolved and Cu^{2+}(aq.) is treated with KI. I_2 liberated required 12.12 mL of 0.1 M $Na_2S_2O_3$ solution for titration. The % Cu in the ore in the ore is:
(A) 2.5% (B) 7%
(C) 5.8% (D) 4.2%

Space for Rough work

Paragraph for Question Nos. 31 and 32

Suppose 50 bacteria are placed in a flask containing nutrients for the bacteria so that they can multiply. A study at 35°C gave the following results:

Time (minutes)	0	15	30	45	60
Number of bacteria	100	200	400	800	1600

31. The rate of the reaction initially is:
(A) 4.62 bacterias per minute
(B) 23.1 bacterias per minute
(C) 23.1 bacterias per sec
(D) 0.231 bacterias per sec

32. The $t_{1/2}$ for the reaction is:
(A) 1.5 sec
(B) 1.5 min
(C) 150 min
(D) 15 min

SECTION – C
(One Integer Value Correct Type)

This section contains **8 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

33. The maximum number of atoms that may lie in the same plane in $(CH_3)_2C = SF_2(CH_3)_2$ is…….

34. H_2S can be dried by how many of the following reagents? Anhydrous $CaCl_2$ conc. H_2SO_4, P_2O_3, KOH solution, Na_2CO_3 solution.

35. Ferrous oxide (FeO) crystal has a cubic structure and each edge of the unit cell is 5.0 Å. Taking density of the oxide as 4.0 g cm^{-3}. The number of Fe^{2+} and O^{2-} ions present in each unit cell is:

36. A closed vessel with rigid walls contains 1 mole of $^{238}_{92}U$ and 1 mole of air at 298 K. Considering complete decay of $^{238}_{92}U$ to $^{206}_{82}Pb$, the ratio of the final pressure to initial pressure the system at 298 K is:

Space for Rough work

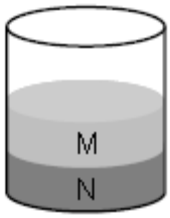
37. For a saturated solution of AgCl at 25°C, specific conductance is 3.41×10^{-6} ohm^{-1} cm^{-1} and that of water used for preparing the solution is $1.60 \times 10^{-6}\,\text{ohm}^{-1}\text{cm}^{-1}$. If $\lambda^{\infty}_{AgCl} = 138.3\ \text{ohm}^{-1}\text{cm}^2\text{eq}^{-1}$, the K_{sp} of AgCl is $0.17 \times 10^{-x}\,M^2$. The value of x is:

38. The half-cell potential of a half cell $A^{(x+n)+}, A^{x+}\,|\,Pt$ were found to be as follows:

% of reduced form	24.4	48.8
Half-cell potential(V)	0.101	0.115

The value of n is:

39. Two liquids 'A' (molecular mass = 20) and 'B' (molecular mass = 40) are partially miscible. When 1 mol of A, 3 mol of B are shaken together and allowed to settle, two layer 'M' and 'N' are formed as shown in diagram.



Layer 'M' contains 0.2 mole fraction of 'A' and layer 'N' contains 0.6 mole fraction of A. Calculate the ratio of masses of layer M to layer N.

40. In an organic compound of molar mass greater than 100 g/mol containing only C, H & N the mass % of C is six times the mass % of H, while the sum of mass % of C & H is 1.5 times the mass % of N. What is $\frac{\text{least molecular mass}}{28}$ of compound?

Space for Rough work

Mathematics

PART – III

SECTION – A
One OR More Than One Choice Type

This section contains **8 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE OR MORE THAN ONE** is/are correct.

41. If the sides a, b & c of ΔABC is such that $\frac{a}{1+\alpha^2\beta^2}=\frac{b}{\alpha^2+\beta^2}=\frac{c}{(1-\alpha^2)(1+\beta^2)}$ then

(A) $A=2\tan^{-1}(\alpha/\beta)$ (B) $\beta=2\tan^{-1}(\alpha\beta)$

(C) area of $\Delta ABC=\frac{\alpha\beta ab}{\alpha^2+\beta^2}$ (D) area of $\Delta ABC=\frac{\alpha\beta bc}{\alpha^2+\beta^2}$

42. If A_i is the area bounded by $|x-a_i|+|y|=b_i,\ i\in N,$ where

$a_{i+1}=a_i+\frac{3}{2}b_i$ & $b_{i+1}=\frac{b_i}{2}, a_1=0, b_1=32,$ then

(A) $A_3=128$ (B) $A_3=256$

(C) $\lim\limits_{n\to\infty}\sum\limits_{i=1}^{n}A_i=\frac{8}{3}(32)^2$ (D) $\lim\limits_{n\to\infty}\sum\limits_{i=1}^{n}A_i=\frac{4}{3}(16)^2$

43. $\bar a$ & $\bar b$ be two vectors such that $|a|=1, |b|=4$ & $\bar a.\bar b=2$. If $\bar c=(2\bar a\times\bar b)-3\bar b,$ then which of the following is/are correct?

(A) $\bar b.\bar c=48$ (B) $\bar b.\bar c=-48$

(C) Angle between $\bar b$ & $\bar c$ is $\frac{5\pi}{6}$ (D) Angle between $\bar b$ & $\bar c$ is $\frac{\pi}{6}$.

44. $f(x)$ is defined for $x\ge 0$ & has a continuous derivative. It satisfies $f(0)=1,\ f'(0)=0$ & $(1+f(x))f''(x)=1+x$. The values f(1) can't take is/are

(A) 2 (B) 1.75
(C) 1.5 (D) 1.35

Space for Rough work

45. If $\bar{a}$ & $\bar{b}$ are any two unit vectors then possible integers in the range of $\frac{3\left|\bar{a}+\bar{b}\right|}{2}+2\left|\bar{a}-\bar{b}\right|$ is

(A) 2 (B) 3
(C) 4 (D) 5

46. If $f(k)=\sum_{r=1}^{k}\frac{1}{r}$ and $\sum_{r=1}^{2017}f(r)=a\,f(b)+c$ then

(A) b = c (B) $a-b=1$
(C) $b+c=0$ (D) $a+c=0$

47. If $\alpha(\theta)\in R$ & $\beta(\theta),\theta\in R-\left\{2n\pi-\frac{\pi}{2},n\in I\right\}$ are functions satisfying $(1+x)\sin^2\theta-\left(1+x^2\right)\sin\theta+\left(x-x^2\right)=0$ then which of the following is/are correct?

(A) $\lim_{\theta\to0^+}\left\{\left(\alpha(\theta)\right)^{\frac{1}{\sin\theta}}+\left(\beta(\theta)\right)^{\frac{1}{\sin\theta}}\right\}=\frac{1}{e^2}$ (B) $\ln\left(\beta(\theta)\right)$ is an odd fn

(C) $\lim_{\theta\to0}\left(\sum_{r=1}^{n}r^{\frac{1}{\alpha^2(\theta)}}\right)^{\alpha^2(\theta)}=n,n\in N,h\geq2$ (D) $\lim_{\theta\to\pi/2}\frac{\alpha(\theta)-\left(\alpha(\theta)\right)^{\alpha(\theta)}}{1-\alpha(\theta)+\ln\left(\alpha(\theta)\right)}=2$

48. The product of 13×5 matrix & 5×13 matrix contains a variable entry x in exactly 2 places. If D(x) is the determinant of the matrix product such that $D(0)=1,D(-1)=1\&D(2)=7$ which of the following is/are correct?

(A) $D(-2)=3$ (B) $D(1)=3$
(C) $D(-3)=7$ (D) $D(1)=2$

Space for Rough work

Comprehension Type

This section contains **2 groups of questions.** Each group has 2 multiple choice question based on a paragraph. Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which only **ONE** is correct.

Paragraph for Question Nos. 49 and 50

Let ABCD is a rectangle with AB = a & BC = b & circle is drawn passing through A & B and touching side CD. Another circle is drawn passing though B & C and touching side AD. Let r_1 & r_2 be the radii of these two circle respectively.

49. $\frac{r_1}{r_2}$ equals

(A) $\frac{a}{b}\left(\frac{4b^2+a^2}{4a^2+b^2}\right)$ (B) $\frac{b}{a}\left(\frac{4a^2+b^2}{4b^2+a^2}\right)$

(C) $\frac{a}{b}\left(\frac{4b^2-a^2}{4a^2-b^2}\right)$ (D) $\frac{b}{a}\left(\frac{a^2-4b^2}{4a^2-b^2}\right)$

50. Minimum value of (r_1+r_2) equals

(A) $\frac{5}{8}(a-b)$ (B) $\frac{5}{8}(a+b)$

(C) $\frac{3}{8}(a-b)$ (D) $\frac{3}{8}(a+b)$

Paragraph for Question Nos. 51 and 52

If $f:R\to[0,\infty)$ be a function satisfying the property $f(x+y)-f(x-y)=f(x)\left[f(y)-f(-y)\right]\ \forall y,x\in R$, $f'(0)=\ln a,\ f(0)=1$

51. f(x) is

(A) e^x (B) 2 ln x

(C) 4 x (D) a^x

Space for Rough work

52. The solution of differential equation $\frac{dy}{dx}=\frac{\left(\log_a f(x)+\log_a f(y)\right)^2}{\left[\log_a f(x)+2\right]\left[\log_a f(y)-2\right]}$ is

(A) $\frac{y-2}{x+2}-\frac{1}{2}\ln\left|1+2\left(\frac{y-2}{x+2}\right)\right|=2\ln(x+2)+c$ (B) $\frac{y+2}{y-2}-\ln\left|1+\frac{y+2}{x-2}\right|=c$

(C) $(x+2)(y-2)=\ln\left(\frac{x+2}{y-2}\right)+c$ (D) None of these

SECTION – C
(One Integer Value Correct Type)

This section contains **8 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

53. If $P(x)=x^6-x^5-x^3-x^2-x$ & $\alpha,\beta,\gamma,\delta$ are the roots of the equation $Q(x)=x^4-x^3-x^2-1=0$ Then $P(\alpha)+P(\beta)+P(\gamma)+P(\delta)=$

54. Let x be chosen at random from the interval $(0,1)$. The probability that $[\log_{10}4x]=[\log_{10}x]$ is k/12 ($[.]$ denotes G.I.F), then k is

55. Let f(x) be a differentiable function on [0, 8] such that $f(1)=3,\ f(2)=1/2, f(3)=4, f(4)=-2, f(5)=6$ $f(6)=1/3, f(7)=-1/4$. Then the minimum no. of points of interaction of the curve $y=f'(x)$ and $y=f'(x)\left(f(x)\right)^2$ is K then $K-5=$

56. Consider continuous function f on the interval [0, 1] which satisfy the following two condition

(i) $f(x)\le\sqrt{5}\ \forall x\in[0,1]$ (ii) $f(x)\le\frac{2}{x}\forall x\in\left[\frac{1}{2},1\right]$

Then the smallest real number α each that the inequality $\int_0^1 f(x)dx\le\alpha$ holds for any such f is $2+2\ln\left(\frac{\sqrt{k}}{2}\right)$ then k is

Space for Rough work

57. 40 slips are placed in a box, each bearing a number 1 to 10 with each number entered on 4 slips. 4 slips are drawn from the box at random without replacement. The probability that two of the slips bear a number 'a' & other two bear a number 'b' $(\neq a)$ is $\frac{k}{9}\times\frac{^{10}C_2}{^{40}C_4}$ where k = ...

58. Suppose f is continuous on $[a,b]$ & $f(a)=f(b)=0$ & $\int_a^b f^2(x)dx=1$ then the minimum value of $\int_a^b (f'(x))^2 dx \int_a^b x^2 f^2(x)dx$ is k, then 8k is

59. The number of solutions of the equation $\left|\sin\frac{\pi x}{2}+\cos\frac{\pi x}{2}\right|=\sqrt{\left(\ln|x|^3+1\right)}$ is

60. Using the digits 0, 1, 2, 3 & 4, the number of ten digit sequences can be written so that the difference between any two consecutive digits is 1, is equal to k, then $\frac{k}{72}$ is equal to

Space for Rough work